

ISSUE 01 | SEPTEMBER 2026

SOIT RESEARCH + INNOVATION NEWSLETTER

Research. Innovation. Impact.

School of Information Technology | STADIO Higher Education

Inside this issue

- 01 SOIT Research Output Snapshot
- 02 Online Research Day Webinar Recap
- 03 Innovation: Assessment Design in the AI Era
- 04 SOIT Research Agenda 2026-2028
- 05 SAICTA Ecosystem Opportunity + Vision 2030
- 06 2027 Research Launch Year: Research Calendar

RESEARCH OUTPUT SNAPSHOT

AS AT 6 SEPTEMBER 2026 | TRACKED SOIT SUBMISSIONS AND OUTPUTS

21

Tracked records

15

Conference records

6

Journal records

12

Accepted outputs

From activity to impact

SOIT guidance favours a quality-first trajectory: credible peer-reviewed outputs, evidence-ready publication records and a sustained 2-3 year pathway toward stronger journals and collaboration.



PIPELINE STATUS

57% accepted. Another 29% is under review.

The active portfolio spans conference and journal routes, with visible clusters in student success, AI, cybersecurity, gamification, AutoML and applied analytics.



In development 2

Submitted 1

Operational priority

Turn every accepted output into a complete evidence pack: acceptance, peer-review evidence, proceedings or journal details, invoices/fees where applicable, and ethics/gatekeeper documentation.

What the numbers say

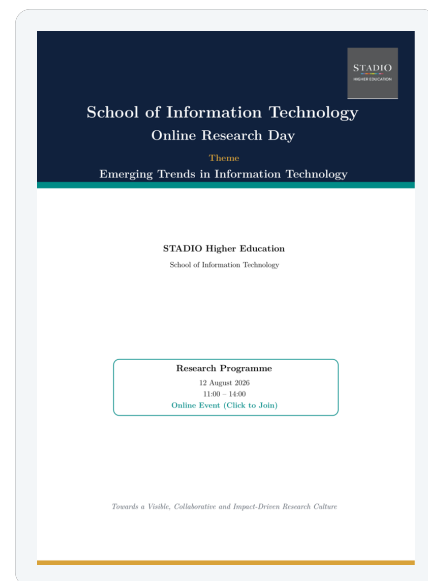
- **7 staff contributors** are represented in the active reporting dataset.
- **10 conference outputs + 2 journal outputs** are recorded as accepted.
- **6 additional outputs** are under review, strengthening the near-term pipeline.
- Recurring themes include **student retention**, **AI in higher education**, and **AI/cybersecurity**.

Quality signal

The objective is not simply more outputs; it is a reliable research pipeline in which each output is visible, defensible and claim-ready.

RESEARCH DAY WEBINAR RECAP

12 AUGUST 2026 | EMERGING TRENDS IN INFORMATION TECHNOLOGY



A VISIBLE, COLLABORATIVE AND IMPACT-DRIVEN RESEARCH CULTURE

Three hours. One research community. Seven research-facing sessions.

The SOIT Online Research Day created a focused platform for staff, postgraduate researchers and invited scholars to share emerging work, receive scholarly feedback, identify publication opportunities and strengthen collaboration across the School.

PROGRAMME AT A GLANCE

11:00	Opening + welcome	Dr Michael Ajayi: Research Day purpose and objectives.
11:05	Keynote	Dr Tshepo Kukuni: Emerging Trends in Information Technology Research.
11:30-13:10	Six research talks	Scientific computing, cybersecurity, AutoML, recruitment analytics, at-risk analytics and digital health.
13:10	Research development	Dr Belinda Ndlovu: practical writing for journals, conferences and proposals.
13:40	Collaboration discussion	Publication pathways, mentorship and the emerging SOIT research agenda.
13:55	Closing	Dr Jan Mentz, Head of School.

AI + Scientific Computing

AutoML, physics-informed neural networks, algorithm configuration and generation.

Cybersecurity + Digital Trust

Blockchain-enabled DDoS mitigation, SDN, resilient systems and network security.

Education + Analytics

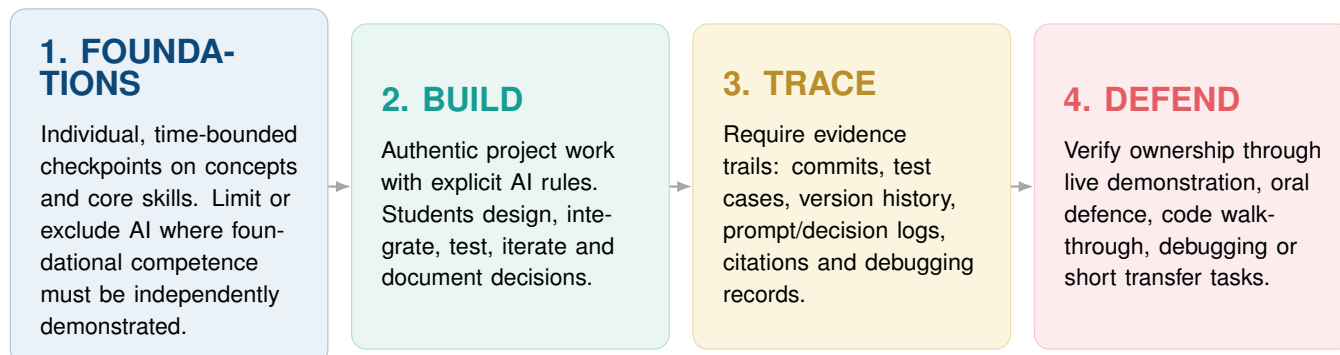
Recruitment analytics, at-risk identification, learning analytics and research skills development.

Next move from the webinar

Convert conversations into concrete writing groups, co-authorship, supervision links and publication-ready manuscripts. The event's value is realised when visibility becomes output.

The design question is no longer “How do we stop AI?” It is “How do we require credible evidence of learning when AI is available?”

A FOUR-STAGE ASSESSMENT ARCHITECTURE



Five design rules

- Assess **process + product**, not only the final artefact.
- Make AI use **explicit, bounded and disclosable**.
- Protect fundamentals with **individual checkpoints**.
- Require **testing, reproducibility and evidence trails**.
- Make students **explain and transfer** what they built.

ILLUSTRATIVE WEIGHTING - NOT POLICY

20%	45%	20%	15%
Foundation check	Individual concepts / fundamentals		
Guided project	Authentic build with bounded AI use		
Controlled validation	Unseen task, test or debugging		
Viva + reflection	Explain, justify and defend decisions		

Why this belongs in SOIT

The School's 2026-2028 agenda explicitly includes technology-enhanced pedagogy, digital curriculum design, AI-driven assessment and personalised learning. The proposed research lab also identifies AI-assisted programming education and AI dependency as research themes.

What this could look like in a programming module

Concept check → **GitHub-based guided project** → **test cases + commit evidence** → **live code walkthrough / debugging task**.
The final mark is supported by multiple forms of evidence rather than a single finished artefact.

SOIT RESEARCH AGENDA

FOCUS AREAS 2026-2028 | APPLIED, SOCIALLY RELEVANT, IMPACT-DRIVEN

The School's research agenda is organised around three complementary areas aligned to STADIO's RRITE values: **Relevant, Responsive, Innovative, Transformative and Effective**.

01

Applied Artificial Intelligence for Industry and Society

Responsible and explainable AI; AI for social good; automation and robotics; AI-enabled productivity and innovation for SMEs.

02

Use of ICT in Education and Digital Innovation

Technology-enhanced pedagogy; digital curriculum; AI-driven assessment; personalised learning; virtual labs; micro-credentials; digital inclusion.

03

Data-Driven Decision Support and Analytics

Decision-support systems; predictive analytics; dashboards; visualisation; policy and risk models; data ethics, privacy and governance.

WHAT IS ALREADY VISIBLE IN THE 2026 PIPELINE

Student retention + early warning

AI in higher education

AI + cybersecurity

AutoML + PINNs

Gamification

UX + applied information systems

From focus area to research output

The strongest opportunity is to connect existing teaching, supervision and applied projects to one of the three focus areas, then move each project through a disciplined research pipeline: clear question, appropriate ethics/permission, credible method, peer-reviewed output and an evidence-ready record.

Research governance reminder

Ethics approval and institutional/gatekeeper permission are distinct. Where STADIO people, data or systems are involved, obtain the appropriate approvals **before** data collection.

Quality over volume

Target credible peer-reviewed outputs, avoid predatory outlets, and keep a clean evidence trail from proposal to publication and claim.

INDUSTRY ECOSYSTEM SPOTLIGHT

SAICTA | CONNECTING EDUCATION, TALENT AND SOUTH AFRICA'S ICT ECOSYSTEM

WHY SAICTA MATTERS TO SOIT

A potential bridge between academic capability and the ICT ecosystem.

SAICTA positions itself as a national ICT association connecting companies and professionals across sectors, with emphasis on professional development, networking, industry visibility and policy influence. For SOIT, the value proposition is not a logo on a page - it is access to conversations, opportunities and industry signals.

Education
↔
Industry

Talent
↔
Opportunity

Innovation
↔
Impact

SAICTA platforms relevant to a School of IT

- **Industry Connect:** a channel linking ICT employers and job seekers.
- **Professional development:** training and CPD-oriented activities.
- **Digital Dawn + Masterclasses:** trend, networking and industry-insight sessions.
- **ICT Excellence Awards:** visibility for innovation and leadership.
- **Thought leadership + policy advocacy:** a broader sector voice.

Potential SOIT engagement

1. Define an institutional engagement pathway.
2. Use events and masterclasses for curriculum and staff-development intelligence.
3. Link students and alumni to Industry Connect opportunities.
4. Position SOIT research, prototypes and thought leadership in relevant channels.
5. Explore speakers, challenge statements, datasets and applied research collaboration.

Strategic fit

SAICTA's emphasis on talent, networking, professional development and ICT-sector visibility aligns naturally with SOIT's goals for employability, partnerships, research visibility and digital innovation.

[VISIT SAICTA.ORG](https://www.saicta.org)

VISION + FUTURE MILESTONES

SOIT APPLIED AI, ANALYTICS AND DIGITAL INNOVATION LAB

LAB VISION

To establish a leading Applied AI, Analytics and Digital Innovation Lab that advances research, innovation, student success and industry collaboration through evidence-based computing solutions.

FOUR RESEARCH PILLARS

P1	Learning Analytics, Student Success, Educational AI
P2	Human-Centred AI, Inclusive Design, Accessibility
P3	Career Intelligence, Employability Analytics, Recommendation Systems
P4	Cybersecurity, Data Science, Digital Innovation

2027-2030 OUTPUT AMBITION

Journal papers	15+
Conference papers	20+
Honours projects	25+
Masters studies	10+
Industry partnerships	5+
Software prototypes	8+

PROPOSED INNOVATION PROJECTS

AI Academic Advisor

Student risk monitoring, intervention recommendations and academic planning.

Document Fluency AI

Policy interpretation, instructional guidance and intelligent decision support.

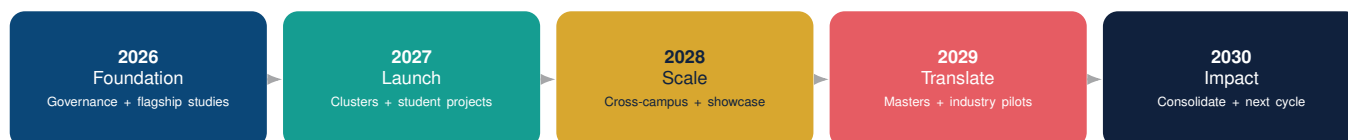
AI Research Assistant

Literature analysis, proposal generation and research-gap identification.

Assessment Design in the AI Era

AI-resilient assessment architecture combining foundations, authentic build, evidence trails and controlled defence of learning.

PROPOSED MILESTONE PATHWAY



What success would look like

A visible platform where staff research, postgraduate supervision, student innovation and industry collaboration reinforce one another - producing publications, prototypes, talent and measurable institutional impact.

2027 RESEARCH LAUNCH YEAR

SOIT RESEARCH CALENDAR | PROPOSED MONTHLY TIMELINE

LAUNCH-YEAR INTENT

Build a predictable annual research rhythm that moves ideas into manuscripts, conference submissions, collaborative projects and visible School-level research outputs.

MONTHLY RESEARCH CALENDAR

JAN	Research launch + output mapping
FEB	Research Writing Retreat
MAR	SACLA, SAICSIT 2027 prep: manuscript development
APR	
MAY	SOIT Webinar 1
JUN	
JUL	STADIO Conference 2027 prep: manuscript development
AUG	SOIT Research Day
SEP	SOIT Webinar 2
OCT	Research Lecture and Project Day
NOV	Publications + claim sprint
DEC	Annual review + 2028 pipeline

ANCHOR ACTIVITIES

Writing Retreat

Concentrated manuscript development and writing-group formation.

Webinar 1 + 2

Research methods, ethics, publication development and invited expertise.

Research Day

Internal showcase, feedback, collaboration and publication conversion.

Conference Prep

SACLA 2027, SAICST 2027 and STADIO conference submission readiness.

Research rhythm

Write → Share → Submit → Showcase → Translate

Launch-year success measure

Each major activity should leave a traceable output: a manuscript draft, ethics submission, conference paper, collaboration, prototype, publication record or claim-ready evidence pack.

Planning note: the 2027 calendar is an indicative SOIT research-planning framework for the launch year. Specific webinar, retreat and conference dates should be confirmed once institutional and conference calendars are published.

THE NEXT MOVE

TURN RESEARCH ACTIVITY INTO VISIBLE IMPACT.

The opportunity for SOIT is not only to produce more research. It is to build a repeatable system that connects teaching innovation, credible research design, industry relevance, student development and visible outputs.

Near-term priorities

- Research launch year (2027).
- Move accepted and under-review work through publication and claim pipelines.
- Convert Research Day connections into writing groups, co-authorship and mentorship.
- Pilot AI-resilient assessment designs that generate evidence of genuine learning.
- Translate the Research Lab proposal into a staged implementation roadmap.
- Define high-value industry engagement channels, including potential SAICTA opportunities.

Research. Innovation. Impact.

Credit: Dr Jan Mentz | SOIT - Head of School

Editor: Dr Michael Ajayi | SOIT Research Representative

School of Information Technology | STADIO Higher Education